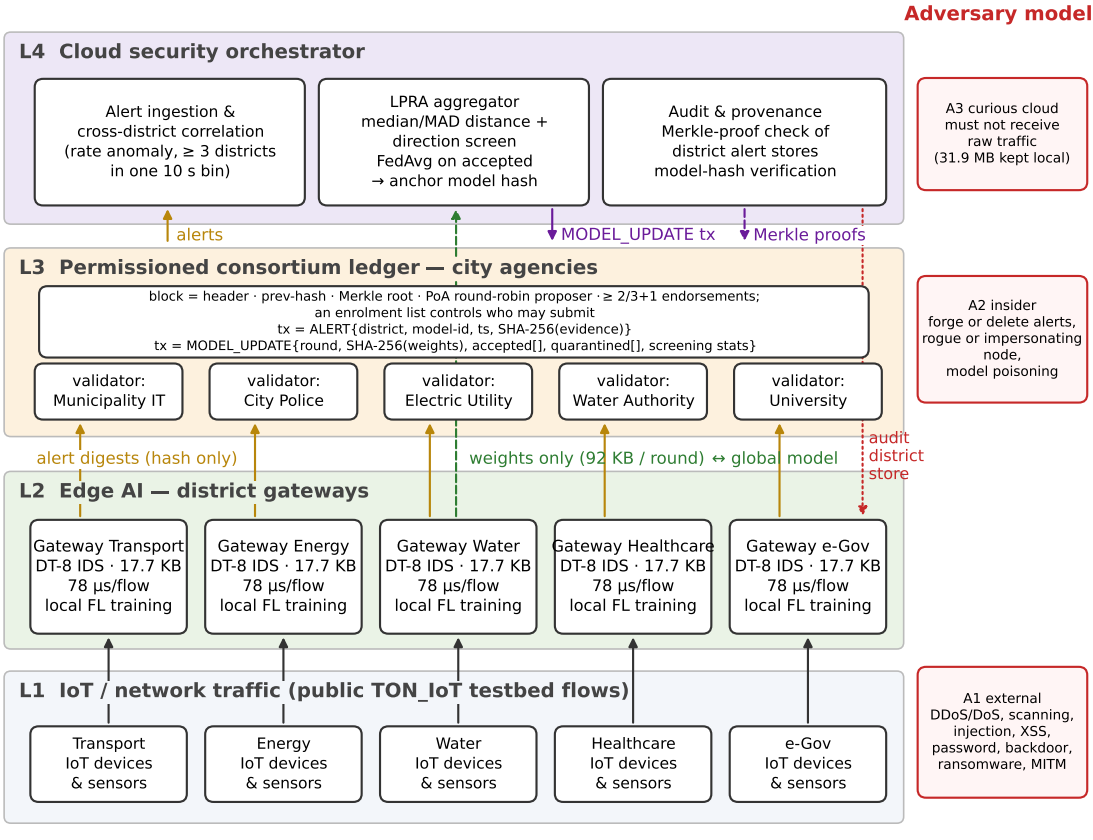
Adversary model
L4 Cloud security orchestrator
Alert ingestion & cross-district correlation (rate anomaly, ≥ 3 districts in one 10 s bin)
LPRA aggregator median/MAD distance + direction screen FedAvg on accepted → anchor model hash
Audit & provenance Merkle-proof check of district alert stores model-hash verification
A3 curious cloud must not receive raw traffic (31.9 MB kept local)
alerts
MODEL_UPDATE tx
Merkle proofs
L3 Permissioned consortium ledger — city agencies
block = header · prev-hash · Merkle root · PoA round-robin proposer · ≥ 2/3+1 endorsements;
an enrolment list controls who may submit
tx = ALERT{district, model-id, ts, SHA-256(evidence)}
tx = MODEL_UPDATE{round, SHA-256(weights), accepted[], quarantined[], screening stats}
A2 insider forge or delete alerts, rogue or impersonating node, model poisoning
validator: Municipality IT
validator: City Police
validator: Electric Utility
validator: Water Authority
validator: University
alert digests (hash only)
weights only (92 KB / round) ↔ global model
audit district store
L2 Edge AI — district gateways
Gateway Transport DT-8 IDS · 17.7 KB 78 µs/flow local FL training
Gateway Energy DT-8 IDS · 17.7 KB 78 µs/flow local FL training
Gateway Water DT-8 IDS · 17.7 KB 78 µs/flow local FL training
Gateway Healthcare DT-8 IDS · 17.7 KB 78 µs/flow local FL training
Gateway e-Gov DT-8 IDS · 17.7 KB 78 µs/flow local FL training
L1 IoT / network traffic (public TON_IoT testbed flows)
Transport IoT devices & sensors
Energy IoT devices & sensors
Water IoT devices & sensors
Healthcare IoT devices & sensors
e-Gov IoT devices & sensors
A1 external DDoS/DoS, scanning, injection, XSS, password, backdoor, ransomware, MITM